

Time : (3 Hours)

Marks: 80

- N.B: 1) Question **number 1** is compulsory.
 2) Attempt **any three** out of the remaining.
 3) Assume suitable data if **necessary** and justify the assumptions.
 4) Figures to the **right** indicate full marks.

Q 1

- A Explain how replication helps in achieving fault tolerance. [5]
 B List and describe five challenges faced in distributed systems. [5]
 C Elaborate key Features of Global Scheduling algorithm. [5]
 D Explain synchronization in DFS with its challenges. [5]

Q 2

- A Explain RPC in distributed systems. [10]
 B Analyze the scalability of the Google File System (GFS). [10]

Q 3

- A Compare two election algorithms and recommend the most efficient one for large-scale distributed systems. [10]
 B Explain the concept and necessity of logical clocks in distributed systems, and describe Lamport's algorithm. [10]

Q 4

- A Describe how Raymond's algorithm ensures mutual exclusion. [10]
 B Explain the working of Distributed File System with its applications. [10]

Q 5

- A Explain how load balancing benefits a distributed system. [10]
 B Explain the Suzuki-Kasami Broadcast Algorithm for mutual exclusion. [10]

Q 6 Write a short note on

- A Code Migration in Distributed System [10]
 B Middleware in Distributed System [10]